

Lorenzo Carpentieri

Curriculum Vitae

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Education

- 2022–2025 **PhD in Computer Science**, *University of Salerno*, Fisciano
Advisors: Biagio Cosenza.
Approximate and energy efficient computing, implementing software techniques to improve performance and energy efficiency in large scale heterogeneous systems. I conducted an in-depth analysis of high-level and low-level GPU programming models, evaluating their expressiveness, performance portability, and ability to approach native low-level APIs on modern SIMD/SIMT accelerators. Designed and implemented domain-specific abstractions and software techniques to enable approximate and energy-aware optimizations on heterogeneous architectures. Investigated compiler-based approaches for performance optimization, including an extensive study of autovectorization capabilities and limitations on RISC-V vector architectures.
- 2020–2022 **Master's Degree in Computer Science**, *University of Salerno*, Fisciano, 110/110 *Cum Laude*
Advisor: Biagio Cosenza
Optimization and porting of the LiGen drug discovery pipeline from NVIDIA CUDA implementation to a portable SYCL implementation.
- 2017–2020 **Bachelor's Degree in Computer Science**, *University of Salerno*, Fisciano, 110/110 *Cum Laude*
Advisor: Amelia Giuseppina Nobile
Performance analysis of tandem queueing networks modeled through birth–death stochastic processes.

Experience

- 2026–Now **Research Scholarship**, *University of Salerno*, Fisciano
Performance and energy characterization of GPU-to-GPU communication on large-scale heterogeneous clusters, using vendor-specific communication libraries such as NCCL, RCCL, and oneCCL.
Detailed achievements:
○ Deployment of the SYnergy energy interface on the Aurora supercomputing system at Argonne, integrated with the GEOPM energy runtime system to support frequency scaling techniques on large-scale in production clusters.
- 2024–2025 **Visiting Research Period**, *Technincal University of Hamburg*, Hamburg
Supervisor: Sohan Lal.
Development of software approximate computing techniques exploiting new architectural features of NVIDIA Hopper and Blackwell GPUs.
Detailed achievements:
○ Developed a hierarchical approximate memoization technique leveraging NVIDIA Thread Block Clustering on NVIDIA Blackwell and H100 GPUs.
○ Achieved up to 2x performance improvement with less than 10% accuracy loss on image processing benchmarks.

- 2022–2024 **Teaching Assistant**, *University of Salerno, Fisciano*
Teaching assistant for Bachelor's and Master's courses in Data Structures, Distributed Programming, and High Performance Computing.
Detailed achievements:
 - Supervised HPC projects and Master's theses involving MPI, OpenMP, SYCL, and RISC-V vectorization.
 - Delivered practical lab sessions on data structures and distributed programming.
- 2021–2022 **Research scholarship**, *University of Salerno, ISIS Lab, Fisciano*
Advisor: Biagio Cosenza.
Performance and energy optimization of the LiGen drug discovery pipeline for the LIGATE EuroHPC project.
Detailed achievements:
 - Porting of the NVIDIA CUDA code to SYCL.
 - Achieved performance portability of 90% on AMD, Intel and NVIDIA GPUs.
 - Implementation of SYnergy a portable energy interface for fine-grained and coarse grained frequency scaling implemented in SYCL by wrapping vendor-specific management libraries such as NVML, ROCm SMI and Level Zero.
 - Energy and performance improvement of up to 20% by designing novel GPU frequency scaling techniques based on machine learning models.

Projects

- 2023–Now **PRIN Project LibreRT: Portable Heterogenous Real-time Programming for the Embedded Computing Continuum**, *University of Salerno, Politecnico di Milano*
Developed SYprox a SYCL library to implement approximate computing techniques on heterogeneous architectures and contributed to the design of SYrtos a SYCL implementation for real-time systems based on pthreads and MIOSEX.
- 2022–2025 **EuroHPC project LIGATE: Ligand Generator and portable drug discovery platform AT Exascale**, *Leader Dompe srl, partners POLIMI, CINECA, KTH, University of Salerno, University of Innsbruck, University of Basel, TU Ostrava, E4, Chelonia, tofmotion*.
Contributed to Work Package 2 across Deliverables D2.1–D2.4, covering the SYCL-based acceleration of the LiGen application redesigning CUDA kernels with SYCL to ensure portability across AMD, Intel and NVIDIA GPUs. Activities also included the performance optimization for the SYCL codebase by leveraging SYCL2020 features and the implementation of new portable frequency scaling techniques for energy efficient computing.

Publications

- IWOCL'23 **Carpentieri, Lorenzo**, and Biagio Cosenza. 'Towards a SYCL API for approximate computing.' Proceedings of the 2023 International Workshop on OpenCL. 2023.
- SC'23 Fan, K., D'Antonio, M., **Carpentieri, L.**, Cosenza, B., Ficarelli, F., Cesarini, D. 'Synergy: Fine-grained energy-efficient heterogeneous computing for scalable energy saving.' In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis.

- SHiPS'23 **Carpentieri, L.**, et al. "Domain-specific energy modeling for drug discovery and magnetohydrodynamics applications." Proceedings of the SC'23 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis. 2023.
- FGCS'24 Crisci, L., **Carpentieri, L.**, Cosenza, B., Accordi, G., Gadioli, D., Vitali, E., Palermo, G., Beccari, A. R. (2024). Enabling performance portability on the LiGen drug discovery pipeline. Future Generation Computer Systems, 158, 44-59.
- IWOCL'24 Accordi, G., Gadioli, D., Palermo, G., Crisci, L., **Carpentieri, L.**, Cosenza, B., Beccari, A. R. (2024, April). Unlocking performance portability on LUMI-G supercomputer: A virtual screening case study. In Proceedings of the 12th International Workshop on OpenCL and SYCL.
- IWOCL'24 Crisci, L., **Carpentieri, L.**, Thoman, P., Alpay, A., Heuveline, V., Cosenza, B. (2024, April). Sycl-bench 2020: Benchmarking sycl 2020 on amd, intel, and nvidia gpus. In Proceedings of the 12th International Workshop on OpenCL and SYCL.
- PDP'25 **Carpentieri, L.**, VazirPanah, M., Cosenza, B. (2025, March). A Performance Analysis of Autovectorization on RVV RISC-V Boards. In 2025 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP).
- IPDPS'25 **Carpentieri, L.**, De Caro, A., Beni, M. S., Fan, K., Cosenza, B. (2025, June). Phase-based Frequency Scaling for Energy-efficient Heterogeneous Computing. In 2025 IEEE International Parallel and Distributed Processing Symposium (IPDPS).
- ICS'25 **Carpentieri, L.**, Cosenza, B. (2025, June). SYprox: Combining Host and Device Perforation with Mixed Precision Approximation on Heterogeneous Architectures. In Proceedings of the 39th ACM International Conference on Supercomputing.
- PDP'26 **Carpentieri, L.**, Cosenza, B. (2026, January). Optimizing the LiGen Drug Discovery Pipeline for Intel Max GPUs. In 2026 34rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP).

Technical Skills

Programming Languages	C, C++, Java, Java EE, Python, Bash, CMake, R.
Programming Models	SYCL, CUDA, HIP, OpenCL, Kokkos, OpenMP, MPI (NCCL, RCCL and oneCCL), RISC-V RVV, Intel AVX, ARM SVE.
Job Schedulers	SLURM, PBS.
Tools	Intel VTune, NVIDIA Nsight Systems/Compute, Unitrace, NVML, ROCm SMI, Level Zero, Git, Docker.

Languages

Italian	Native
English	Fluent

Conferences given in English